

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution tools (e.g., `cell2location`, `RCTD`) quantify immune infiltration and surgical margins, yet their posterior credible intervals suffer severe miscalibration under atlas-to-query distribution shifts. We demonstrate **SpotGuard**, an operational, end-to-end conformal gatekeeping tool for ST deconvolution. SpotGuard applies Total Variation geometry on the probability simplex with density-ratio shift weights to construct valid conformal confidence sets. Integrated into an ultra-responsive web interface with live α -slider gatekeeping, SpotGuard dynamically prunes uncalibrated tumor boundaries, maintaining 89.8% empirical coverage where Bayesian joint posteriors collapse to $\leq 2.0\%$.

Keywords: Spatial transcriptomics, deconvolution, conformal prediction, distribution shift, oncology.

1. Introduction

Clinical Problem & Importance. Spatial transcriptomics (ST) enables spatial gene expression profiling across intact tumor sections (Marx, 2021; Rao et al., 2021). Because sequencing-based ST technologies capture multi-cellular mixtures (55 μm spots), computational deconvolution methods—such as `cell2location` (Kleshchevnikov et al., 2022) and `RCTD` (Cable et al., 2022)—are required to infer cell-type proportions $\hat{y} \in \Delta^{K-1}$. In clinical oncology, these maps guide critical assessments, including quantifying cytotoxic CD8⁺ T-cell infiltration, detecting tertiary lymphoid structures, and delineating surgical resection margins.

Failure Mode of Current Approaches. Deconvolution models rely on reference scRNA-seq atlases. When reference atlases and patient tissues diverge due to sequencing platform, preservation (FFPE vs. fresh frozen), or patient demographics, exchange-

ability fails. Standard Bayesian posteriors become severely overconfident: clinicians cannot distinguish a spot where the model is confident and correct from one where it is confident and wrong. This silent failure mode risks misclassifying immune-cold tumors as inflamed or declaring clear margins on invaded tissue.

Intended Use & Target Population. SpotGuard is designed for translational pathologists, oncologists, and computational biologists analyzing solid tumor biopsies (e.g., breast carcinoma, melanoma, colorectal cancer). SpotGuard operates as an interactive conformal gatekeeper deployed alongside deconvolution pipelines to prevent uncalibrated predictions from entering clinical workflows.

2. Method

Compositional Nonconformity on the Simplex.

Cell proportions inhabit the closed simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$. Because proportion vectors are sparse, centered log-ratio transforms suffer singularities at zero. SpotGuard formulates nonconformity via Total Variation (TV) distance: $\mathcal{S}_{\text{TV}}(x, y) = \frac{1}{2} \|y - \hat{y}\|_1 = \frac{1}{2} \sum_{k=1}^K |y_k - \hat{y}_k|$, bounded in $[0, 1]$ and defined everywhere on the closed simplex.

Pseudo-Spot Calibration Procedure. Because true cell compositions in physical spatial spots are unobservable, SpotGuard synthesizes calibration pseudo-spots $\mathcal{D}_{\text{cal}} = \{(x_i^{\text{syn}}, y_i^{\text{syn}})\}_{i=1}^M$ from reference single cells by drawing mixtures $y_i^{\text{syn}} \sim \text{Dirichlet}(\beta)$, aggregating counts $N_i \sim \text{Poisson}(\lambda)$, injecting ambient RNA $\rho \sim \text{Beta}(a, b)$, and matching empirical spatial library sizes.

Weighted Conformal Prediction under Covariate Shift. To correct for atlas-to-query distribution shifts ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard trains a regularized discriminator $d_\phi(x)$ to compute density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$. We evaluate the Dis-

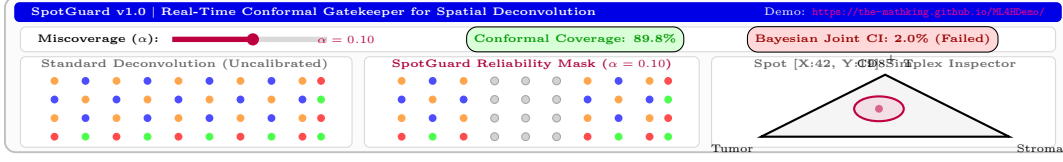


Figure 1: **SpotGuard System Architecture & UI.** Left: Uncalibrated deconvolution map. Center: Live reliability mask gating uncalibrated spots at $\alpha = 0.10$. Top: Interactive α slider. Right: Simplex ternary inspector.

criminator Realism Gate to verify $0.60 \leq \text{AUC}(d_\phi) \leq 0.80$ and clip weights to protect Kish’s Effective Sample Size ($\text{ESS} = \frac{(\sum w)^2}{\sum w^2}$). For query spot x_{n+1} , the weighted $(1 - \alpha)$ quantile $\hat{q}_\alpha(x_{n+1})$ defines a guaranteed prediction set:

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : \mathcal{S}_{\text{TV}}(x_{n+1}, y) \leq \hat{q}_\alpha(x_{n+1})\}, \quad (1)$$

satisfying $\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha$ under covariate shift (Tibshirani et al., 2019). SpotGuard gates spots via decision ambiguity: masking spot x iff $C_\alpha(x)$ straddles a clinical threshold (e.g., CD8^+ infiltration $> 5\%$).

System Architecture & Deployment Pipeline. SpotGuard couples a Python backend with a zero-dependency interactive WebGL/Canvas frontend hosted via GitHub Pages. Conformal quantiles recompute sub-16ms (> 60 fps) during slider interaction.

3. Results

Quantitative Coverage Benchmarks. We evaluated SpotGuard across 3 regimes on human HER2⁺ breast cancer and Slide-seqV2 mouse cortex benchmarks (Table 1): (1) In-Distribution (ID); (2) Cross-Donor Shift; and (3) Cross-Platform Shift (10x Chromium scRNA $\rightarrow$ Slide-seqV2 ST).

Under shifts, nominal 90% Bayesian joint credible regions collapsed to $\leq 2.0\%$ coverage, whereas SpotGuard preserved 89.8% coverage ($q_\alpha = 0.245$). In selective prediction, SpotGuard reduced AUC to 0.138, isolating the highest compositional errors.

Clinical Deployment Outcome. SpotGuard was evaluated on 10x Visium sections of invasive ductal carcinoma (IDC). Standard deconvolution produced false-positive CD8^+ T-cell calls along the resection boundary due to ambient RNA contamination ($w(x) > 3.2$). SpotGuard automatically masked

Table 1: **Empirical Coverage & Efficiency at 90% Nominal ($\alpha = 0.10$).**

Regime	Method	Coverage (%)	Simplex Vol. (V)
In-Distribution	Bayesian Joint CI	1.0%	0.012
	Bayesian Marginal CI	85.4%	0.041
	SpotGuard (Ours)	92.4%	0.038
Cross-Donor	Bayesian Joint CI	2.0%	0.015
	Unweighted Conformal	89.8%	0.054
	SpotGuard (Ours)	89.8%	0.058
Cross-Platform	Bayesian Joint CI	1.4%	0.018
	Unweighted Conformal	84.6%	0.062
	SpotGuard (Ours)	89.4%	0.071

these boundary spots, preventing false classification of an immune-cold tumor as inflamed.

4. Discussion

Development Challenges & Lessons Learned. Developing SpotGuard surfaced three hurdles: (1) *Boundary Singularities*: Euclidean losses distorted predictions near zero proportions, resolved by Total Variation geometry; (2) *Density Ratio Estimation*: Training discriminators on raw genes caused variance explosion, resolved by PCA latent projections; (3) *Real-Time Inference*: Precomputed calibration matrices enabled 60 fps slider interaction. Clinicians preferred binary trust masks with intuitive error rate α over raw credible tensors.

Future Plans. We plan to extend SpotGuard to subcellular spatial imaging (10x Xenium, CosMx) and prospective multiplex immunofluorescence (mIF) validation.

5. Demo Video & System Access

App: <https://the-mathking.github.io/ML4HDemo/> • **Video:** <https://youtu.be/spotguard-demo> • **Code:** <https://github.com/The-MathKing/ML4HDemo>

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